

Repair 1 - JOIN behavior, cardinality and inflation

Scope: SA1-SA2

- State every source grain and desired output grain.
- Prove one-side uniqueness instead of assuming it.
- Count rows before and after a join.
- If a coarse-grain measure repeats, aggregate the many side or compute that measure before multiplication. Never use DISTINCT as a blind fix.

Repair discipline

- Keep the failed/partial SQL as evidence.
- Repair only the weak competency.
- Close answer material before the changed retry.
- Re-run downstream reconciliations after any upstream grain/cardinality repair.

Repair 2 - Strings and dates

Scope: SA3-SA4

- Keep raw and cleaned text side-by-side.
- Standardization does not prove entity identity.
- Use true date values and a chronological month key.
- NULL completion/end dates remain unknown/not-applicable rather than zero elapsed time.

Repair discipline

- Keep the failed/partial SQL as evidence.
- Repair only the weak competency.
- Close answer material before the changed retry.
- Re-run downstream reconciliations after any upstream grain/cardinality repair.

Repair 3 - Subqueries and CTEs

Scope: SA5-SA6

- Run the inner subquery separately and state its result shape.
- Name every CTE for what one row represents.
- Validate counts/totals at each stage before using the next stage.
- Rewrite opaque nesting into inspectable stages when clarity improves.

Repair discipline

- Keep the failed/partial SQL as evidence.
- Repair only the weak competency.
- Close answer material before the changed retry.
- Re-run downstream reconciliations after any upstream grain/cardinality repair.

Repair 4 - Windows, ranking and LAG

Scope: SA7-SA8

- Compare window row count with the intended detail grain.
- Validate partition values against GROUP BY.
- Latest-row selection needs deterministic tie-breaker.
- Percentage change requires an explicit NULL/zero denominator rule.

Repair discipline

- Keep the failed/partial SQL as evidence.
- Repair only the weak competency.
- Close answer material before the changed retry.
- Re-run downstream reconciliations after any upstream grain/cardinality repair.

Repair 5 - UNION, project and Gate defense

Scope: SA9 / M-SQLA / Gate

- UNION ALL keeps meaningful repeated cohort membership; UNION removes exact selected-row duplicates.
- Trace report finding back to query grain and reconciliations.
- Separate observation from hypothesis.
- Gate B/C are explicit remediation full retests, never automatic progression.

Repair discipline

- Keep the failed/partial SQL as evidence.
- Repair only the weak competency.
- Close answer material before the changed retry.
- Re-run downstream reconciliations after any upstream grain/cardinality repair.